INTERVIEW PRESENTATION — TRACK 2, COMMERCIAL SMART FACILITY & SUSTAINABILITY MANAGER

KOHLER AquaGuard AI

Smart Facility & Water Sustainability Manager

PREPARED BY

Deanne Jerusha Fernandes

Interview Case Presentation

TRACK 2 — COMMERCIAL SMART FACILITY & SUSTAINABILITY MANAGER

KOHLER AquaGuard AI

Smart Facility & Water Sustainability Manager

THE PROBLEM

Airports, hospitals and universities generate huge facility data — yet maintenance often stays reactive, not predictive.

- Continuous water leaks go unnoticed
- Cleaning schedules ignore real restroom usage
- Sensor issues undermine monitoring reliability
- Teams can't tell which issue needs attention first

OUR SOLUTION

An intelligent facility-management prototype combining simulated IoT telemetry with explainable AI-style reasoning.

- Detect abnormal water usage
- Identify potential leaks
- Predict hygiene / cleaning needs
- Monitor sensor health
- Prioritize maintenance actions
- Estimate potential water loss

CORE IDEA

Sense ▶ **Detect** ▶ **Understand** ▶ **Predict** ▶ **Act**

AI-Powered Facility Intelligence Pipeline

How AquaGuard turns raw sensor data into prioritized action

1

Sense

Water flow, flush count, occupancy, leak sensor status, sensor health, time since cleaning

2

Detect

Abnormal patterns — e.g. high flow + low occupancy + low flush activity + leak signal

3

Understand

Combines multiple signals; generates risk level, Facility Intelligence Score & explanation

4

Predict

Estimates consequences — potential water loss if the anomaly continues

5

Act

Leak inspection, cleaning recommendation, sensor attention, maintenance ticket

EXAMPLE — RESTROOM B-203

18.8 L/min abnormal flow detected

If the flow continues for 6 hours

≈ 6,768 L

potential additional water loss

Closed-loop approach: IoT Data → AI Reasoning → Prediction → Action

Technology & Working Prototype



Python

Core application & decision logic



Streamlit

Interactive facility dashboard



Pandas

Sensor-data processing & analysis



CSV / IoT

24 simulated restroom sensor records

MULTI-SIGNAL ANOMALY DETECTION

Example — Restroom B-203

Water flow	**18.8 L/min**
Occupancy	**20%**
Flush count	**4**
Leak sensor	**Triggered**

CRITICAL — Potential Continuous Water Leak



Explainable Reasoning

Beyond a bare alert, AquaGuard explains “why was this restroom flagged?” — making it easy for facility teams to understand and act.

INTERACTIVE DASHBOARD

- Facility Overview
- Restroom Investigator
- Predictive Hygiene
- Maintenance Priority
- Alert Center
- Leak Intelligence
- Sensor Health
- Ask AquaGuard

Prototype → Production: current prototype uses simulated telemetry; future deployment can connect the intelligence layer to live IoT sensors, enterprise maintenance systems and AI services.

Why AquaGuard?



Multi-Signal Intelligence

Combines flow + occupancy + flush activity + leak signal + sensor health for a stronger read on conditions.



Facility Intelligence Score

A single 0-100 score gives a quick view of overall condition and flags areas needing attention.



“What Happens If I Ignore It?”

Estimates impact, not just alerts — e.g. ~6,768 L potential loss over 6 hours at B-203.



Predictive Hygiene

Cleaning recommendations weigh occupancy + flush activity + time since cleaning, not fixed schedules.



Action-Oriented AI

Detect → Explain → Prioritize → Generate maintenance action, beyond passive monitoring.

EXPECTED IMPACT



Water Sustainability

Earlier identification of abnormal consumption



Operational Efficiency

Teams focus on higher-priority issues



Better Facility Hygiene

Usage-based cleaning recommendations



Smarter Monitoring

Combines sensor health with facility conditions

Detect › Explain › Prioritize › Act